

9(a)(i)	P pointing to thin filament ; R pointing to thick filament ;	2
9(a)(ii)	A-band – stays the same ; I -band – gets narrower ;	2

Question	Answer	Marks
9(b)(i)	any four from: 1 competes with, acetylcholine / neurotransmitter or acetylcholine cannot bind to receptors or blocks receptors ; 2 Na⁺ channels do not open or no / fewer, sodium ions enter (muscle fibre) ; 3 so, no / less, depolarisation of sarcolemma ; 4 no / fewer, action potentials / impulses (spread across muscle fibre) ; 5 Ca²⁺ (voltage-gated) channels do not open in sarcoplasmic reticulum or no / fewer, Ca²⁺ ions released by sarcoplasmic reticulum ; 6 no / fewer, Ca²⁺ ions bind to troponin or no exposing of binding sites or no / fewer, cross bridges / AW ;	4
9(b)(ii)	any two from: 1 affects, rib muscles / intercostal muscles / diaphragm, so unable to breathe ; 2 affects cardiac muscle so, stops blood circulation / heart failure / attack ; 3 cannot move so unable to escape from predators ; 4 cannot eat so starves ;	2